

DAY 00 / MASTER INDEX

Days 61-77 Foundation Learning Index

A navigational index for the 17 generated Python foundation notes, using the source titles and topic names.

17

DAILY LESSONS

195

PDF PAGES

3

LEARNING WEEKS

RECOMMENDED PATH

1**Developer basics**

Days 61-63: pip, virtual environments, project structure, requirements.txt, and revision.

2**Git and data collection**

Days 64-70: Git, CSV, Excel, JSON, APIs, debugging, and weekly revision.

3**NumPy, Pandas, and data cleaning**

Days 71-77: NumPy, Pandas, cleaning, encoding, scaling, and weekly revision.

WEEKLY RHYTHM

Follow the lessons in numerical order; each revision day connects the preceding topics through combined practice.

WEEK 01 / DAYS 61-63

Developer tools and project structure

Install packages safely, isolate dependencies, organize a Python project, and connect the workflow through revision.

DAY 61	pip, Packages, and Virtual Environments FOCUS pip, packages, and virtual environments PDF Day-61-pip-and-Virtual-Environments.pdf	10 PAGES
DAY 62	Basic Python Project Structure and requirements.txt FOCUS Project folders and requirements.txt PDF Day-62-Project-Structure-and-requirements.txt.pdf	10 PAGES
DAY 63	Weekly Revision - Developer Basics FOCUS Revision of Days 61-62: Developer Basics PDF Day-63-Weekly-Revision.pdf	11 PAGES
MEMORY LINE A virtual environment isolates dependencies; requirements.txt records them; a clear project structure keeps files understandable.		

WEEK 02 / DAYS 64-70

Git and data collection

Track changes, work with common data files, collect API responses, and handle failures clearly.

DAY 64	Git and GitHub Fundamentals FOCUS Git, Repository, Commit, and GitHub PDF Day-64-Git-and-GitHub-Fundamentals.pdf	12 PAGES
DAY 65	CSV Files in Python FOCUS Reading and Writing CSV Data PDF Day-65-Read-and-Write-CSV-Files.pdf	13 PAGES
DAY 66	Basic Excel File Handling in Python FOCUS Excel Workbook and Worksheet Basics PDF Day-66-Excel-Files.pdf	13 PAGES
DAY 67	JSON Data in Python FOCUS JSON and Python Dictionaries PDF Day-67-JSON-Data.pdf	11 PAGES
DAY 68	Basic API Data Collection Python FOCUS API Request and JSON Response PDF Day-68-Basic-API-Data-Collection.pdf	13 PAGES
DAY 69	Basic Debugging and Error Handling for Data Programs FOCUS Debugging file and API programs PDF Day-69-Debugging-Data-Collection-Programs.pdf	10 PAGES
DAY 70	Weekly Revision - Git and Data Collection FOCUS Weekly Revision - Git and Data Collection PDF Day-70-Weekly-Revision.pdf	14 PAGES

MEMORY LINE

Track changes with Git, collect structured data from CSV, Excel, JSON, and APIs, and handle failures clearly.

WEEK 03 / DAYS 71-77

NumPy, Pandas, and data preparation

Move from numerical arrays to labeled tables, inspection, cleaning, encoding, scaling, and weekly revision.

DAY 71	NumPy Fundamentals - Arrays FOCUS NumPy Fundamentals - NumPy Arrays PDF Day-71-NumPy-Arrays.pdf	10 PAGES
DAY 72	NumPy Indexing and Basic Arithmetic FOCUS NumPy indexing, slicing, and arithmetic PDF Day-72-NumPy-Indexing-and-Arithmetic.pdf	9 PAGES
DAY 73	Pandas Fundamentals - DataFrames FOCUS Pandas Series and DataFrame Basics PDF Day-73-Pandas-DataFrames.pdf	10 PAGES
DAY 74	Loading and Inspecting Data with Pandas FOCUS Reading CSV, Excel, and JSON into Pandas DataFrames PDF Day-74-Load-and-Inspect-Data-with-Pandas.pdf	11 PAGES
DAY 75	Basic Data Cleaning with Pandas FOCUS Basic Data Cleaning with Pandas - Missing Values, Duplicates, and Incorrect Data Types PDF Day-75-Missing-Values-Duplicates-and-Data-Types.pdf	15 PAGES
DAY 76	Basic ML Preprocessing - Encoding and Scaling FOCUS Categorical Encoding, Scaling, and Normalization PDF Day-76-Encoding-Scaling-and-Normalization.pdf	10 PAGES
DAY 77	Weekly Revision - NumPy, Pandas and Data Cleaning FOCUS Revision of Days 71-76: NumPy, Pandas, Data Cleaning, Encoding, and Scaling PDF Day-77-Weekly-Revision.pdf	13 PAGES

MEMORY LINE

Use NumPy for arrays, Pandas for labeled tables, inspect before cleaning, and prepare features deliberately.

QUICK LOCATOR

Find the right lesson fast

Use this map when you know the source topic or PDF filename but not the lesson day.

Topic name	Generated PDF	Go to
pip, packages, and virtual environments	Day-61-pip-and-Virtual-Environments.pdf	Day 61
Project folders and requirements.txt	Day-62-Project-Structure-and-requirements.txt.pdf	Day 62
Revision of Days 61-62: Developer Basics	Day-63-Weekly-Revision.pdf	Day 63
Git, Repository, Commit, and GitHub	Day-64-Git-and-GitHub-Fundamentals.pdf	Day 64
Reading and Writing CSV Data	Day-65-Read-and-Write-CSV-Files.pdf	Day 65
Excel Workbook and Worksheet Basics	Day-66-Excel-Files.pdf	Day 66
JSON and Python Dictionaries	Day-67-JSON-Data.pdf	Day 67
API Request and JSON Response	Day-68-Basic-API-Data-Collection.pdf	Day 68
Debugging file and API programs	Day-69-Debugging-Data-Collection-Programs.pdf	Day 69
Weekly Revision - Git and Data Collection	Day-70-Weekly-Revision.pdf	Day 70
NumPy Fundamentals - NumPy Arrays	Day-71-NumPy-Arrays.pdf	Day 71
NumPy indexing, slicing, and arithmetic	Day-72-NumPy-Indexing-and-Arithmetic.pdf	Day 72
Pandas Series and DataFrame Basics	Day-73-Pandas-DataFrames.pdf	Day 73
Reading CSV, Excel, and JSON into Pandas DataFrames	Day-74-Load-and-Inspect-Data-with-Pandas.pdf	Day 74
Basic Data Cleaning with Pandas - Missing Values, Duplicates, and Incorrect Data Types	Day-75-Missing-Values-Duplicates-and-Data-Types.pdf	Day 75
Categorical Encoding, Scaling, and Normalization	Day-76-Encoding-Scaling-and-Normalization.pdf	Day 76
Revision of Days 71-76: NumPy, Pandas, Data Cleaning, Encoding, and Scaling	Day-77-Weekly-Revision.pdf	Day 77

COMPLETION CHECK

Understand	Explain the concept and predict the output before running code.
Build	Write the exercise without copying, using meaningful names and clean indentation.
Test	Try normal, empty, boundary, duplicate, and invalid inputs where relevant.
Reflect	State one mistake to avoid and one place the pattern applies in a real program.

COLLECTION 17 note PDFs in pdfs/

| TOTAL 195 pages

| UPDATED 27 September 2026